

Eco-Loop Building Agents

Autonomous Digital Twin Executive Audit & Performance Report

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Executive Summary

The Eco-Loop Building Agents autonomous control framework was evaluated across a multi-day building simulation. By integrating real-time machine learning predictions, semantic memory retrieval, and Model Context Protocol (MCP) tool execution, the system actively optimized HVAC setpoints and lighting schedules while ensuring strict adherence to occupant thermal comfort boundaries (Fanger PMV).

Key Performance Indicators (KPIs)

Performance Metric	Static Baseline	AI Autonomous Control	Net Impact / Gain
Total Energy Consumed	1,701.65 kWh	14.02 kWh	99.18% Reduction
Peak Demand Power	0.68 kWh	0.13 kWh	80.51% Shaving
Carbon Emissions	644.06 kg CO2	4.45 kg CO2	639.61 kg Saved
Thermal Comfort (PMV)	100.0% Compliant	100.0% Compliant	Avg PMV: -0.25

Autonomous Decision & Audit Trail

During the simulation, the agent executed a total of **10 autonomous tool calls** across the MCP server interface:

- `apply_ecm`: Executed 10 times

Key Analytical Insights

- Exceptional autonomous performance achieved: 99.2% reduction in total energy consumption compared to static ASHRAE baseline.
- Flawless occupant thermal comfort maintained (100.0% compliance within Fanger PMV [-0.5, +0.5] threshold).
- Peak demand shaving active: maximum interval power consumption reduced by 80.5%, lowering demand charges.

Strategic Recommendations for Facility Managers

- Deploy AI Agent setpoint schedules during high grid carbon intensity hours (>500 gCO₂/kWh) for maximum ESG impact.
- Continue periodic retraining of Random Forest and SVR models with live building sensor streams to prevent concept drift.
- Integrate dynamic electricity tariff signaling (Time-of-Use pricing) into the MCP Server for cost-optimized pre-cooling.